

Questions with Answer Keys

MathonGo

Q1: 25 Feb (Shift 2) - Single Correct

If the curve $x^2 + 2y^2 = 2$ intersects the line $x + y = 1$ at two points P and Q , then the angle subtended by the line segment PQ at the origin is:

(1) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$

(2) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$

(3) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{3}\right)$

(4) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{3}\right)$

Q2: 26 Feb (Shift 2) - Numerical

Let L be a common tangent line to the curves $4x^2 + 9y^2 = 36$ and $(2x)^2 + (2y)^2 = 31$. Then the square of the slope of the line L is

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Answer Key

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Q1 (1)

Q2 (3)

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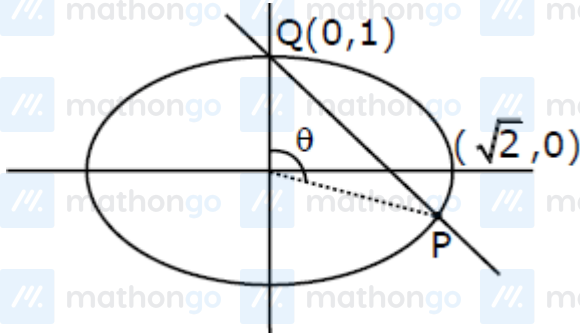
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Q1 (1)

Ellipse $\frac{x^2}{2} + \frac{y^2}{1} = 1$

Line $x + y = 1$



Using homogenisation $x^2 + 2y^2 = 2(1)^2$

$x^2 + 2y^2 = 2(x + y)^2$

$x^2 + 2y^2 = 2x^2 + 2y^2 + 4xy$

$x^2 + 4xy = 0$

for $ax^2 + 2hxy + by^2 = 0$

$\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a+b} \right|$

$\tan \theta = \left| \frac{2\sqrt{(2)^2 - 0}}{1+0} \right|$

$\tan \theta = -4$

$\cot \theta = -\frac{1}{4}$

$\theta = \cot^{-1}\left(-\frac{1}{4}\right)$

$\theta = \pi - \cot^{-1}\left(\frac{1}{4}\right)$

$\theta = \pi - \left(\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)\right)$

$\theta = \frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$

Q2 (3)

$E : \frac{x^2}{9} + \frac{y^2}{4} = 1$ $C : x^2 + y^2 = \frac{31}{4}$

equation of tangent to ellipse is $y = mx \pm \sqrt{9m^2 + 4}$

equation of tangent to circle is

Hints and Solutions

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$$y = mx \pm \sqrt{\frac{31}{4}m^2 + \frac{31}{4}} \dots (ii)$$

Comparing equation (i) \& (ii) $9m^2 + 4 = \frac{31}{4}m^2 + \frac{31}{4}$

$$\Rightarrow 36m^2 + 16 = 31m^2 + 31$$

$$\Rightarrow 5m^2 = 15$$

$$\Rightarrow m^2 = 3$$